NARAYANA ENGINEERING COLLEGE :: GUDUR

DEPARTMENT OF ECE

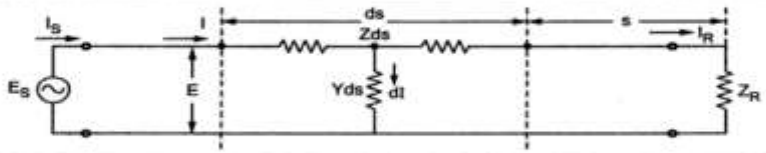
CourseDetails							
Class	:	**II -B.Tech**	**Semester**	**II**	**Year**	:	**2024-25**
CourseTitle	:	**EMTL**	**CourseCode**	: **23EC2005**	**Credits**	:	**3**
Program/Dept.	:	**ECE**	**Section**	: **A, B, C, EVT & ECA**	**Batch**	:	**2023- 2027**
Regulation	:	**NECR-23**	**Faculty**	**:Dr. B. Malakonda Reddy**			

Question Bank /Topic Learning Outcomes

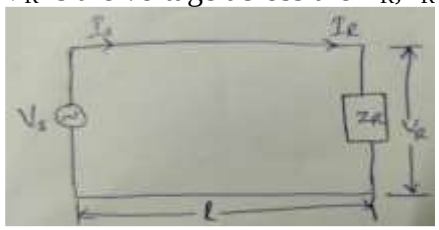
(Table format as given below is suitable for Theory, MCQ, Lab, and Tutorial. Columns can be added or deleted. No need to take printouts)

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
1	**Derive the general solution of a transmission line** The current and voltage varies from point to point along the transmission line. Line equation represents the current and voltage equations at some point on the line. The various notations used in this derivation are R = Series Resistance Ohms per unit length (in Ω /m) L = Series Inductance, in H/m C = Shunt Capacitance, in F/m G = Shunt Conductance, in ℧ /m ωL = Series Reactance ωC = Shunt Susceptance Z = R + j ω L Series Impedance, Y = G + j ω C Shunt Admittance,	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	S = Distance measured from receiving end I = Current in the line at any point V = Voltage between the conductors at any point L = Length of the line This incremental section is of length of ds and carries a current I. The series line impedance being Z ohms and	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	The elemental voltage drop in the length ds is, $dE = I Z ds$ $\therefore \quad \frac{dE}{ds} = I Z$ ... (1) The leakage current flowing through shunt admittance from one conductor to other is given by, $dI = EY ds$ $\therefore \quad \frac{dI}{ds} = EY$ ... (2) Differentiating equations (1) and (2) with respect to S we get $\frac{d^2E}{ds^2} = Z\frac{dI}{ds}$ and $\frac{d^2I}{ds^2} = Y\frac{dE}{ds}$ This is because both E and I are functions of S. $\frac{d^2E}{ds^2} = ZEY$ ... (3) and $\frac{d^2I}{ds^2} = YIZ$ ... (4) The equations (3) and (4) are the second order differential equations describing the transmission line having distributed constants, all along its length. It is necessary to solve these equations to obtain the expressions of E and I.  Fig. 11.17 Infinitesimal T section of a long line Replace the operator d/dS by m hence we get, $(m^2 - ZY)E = 0$ but $E \neq 0$ $\therefore \quad m = \pm\sqrt{ZY}$ ... (5)	

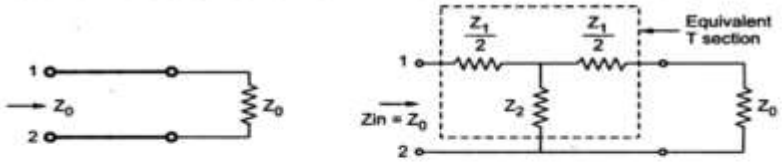
Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Same result is true for the current equation. So there exists two solutions for positive sign of m and negative sign of m. The general solution of the equations for E and I are, $E = A e^{\sqrt{ZY}\, s} + B e^{-\sqrt{ZY}\, s}$ ... (6) $I = C e^{\sqrt{ZY}\, s} + D e^{-\sqrt{ZY}\, s}$ ... (7) where A, B, C and D are arbitrary constants of integration. It is now necessary to obtain the values of A, B, C and D. As distance is measured from the receiving end, s = 0 indicates the receiving end. $E = E_R$ and $I = I_R$ ... at S = 0 Substituting in the solution, $E_R = A + B$ ... (8 (a)) $I_R = C + D$ ... (8(b)) Same condition can be used in the equations obtained by differentiating the equations (6) and (7) with respect to s. $\frac{dE}{ds} = A\sqrt{ZY}\, e^{\sqrt{ZY}\, s} + B\left(-\sqrt{ZY}\right) e^{-\sqrt{ZY}\, s}$ and $\frac{dI}{ds} = C\sqrt{ZY}\, e^{\sqrt{ZY}\, s} + D\left(-\sqrt{ZY}\right) e^{-\sqrt{ZY}\, s}$ But $\frac{dE}{ds} = IZ$ and $\frac{dI}{ds} = EY$ $\therefore$ $IZ = A\sqrt{ZY}\, e^{\sqrt{ZY}\, s} - B\sqrt{ZY}\, e^{-\sqrt{ZY}\, s}$ ... (9) and $EY = C\sqrt{ZY}\, e^{\sqrt{ZY}\, s} - D\sqrt{ZY}\, e^{-\sqrt{ZY}\, s}$ ... (10) $I = \frac{A}{Z}\sqrt{ZY}\, e^{\sqrt{ZY}\, s} - \frac{B}{Z}\sqrt{ZY}\, e^{-\sqrt{ZY}\, s}$ i.e. $I = A\sqrt{\frac{Y}{Z}}\, e^{\sqrt{ZY}\, s} - B\sqrt{\frac{Y}{Z}}\, e^{-\sqrt{ZY}\, s}$ ... (11) And $E = C\sqrt{\frac{Z}{Y}}\, e^{\sqrt{ZY}\, s} - D\sqrt{\frac{Z}{Y}}\, e^{-\sqrt{ZY}\, s}$ ... (12) The above equations are very useful form for the voltage and current at any point on a transmission line. After simplifying the above equations we get the final and very useful form of equations for voltage and current at any point on a k=line, and are solutions to the	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	wave equation. $E = E_R \cosh\sqrt{ZY}s + I_R Z_0 \sinh\sqrt{ZY}s$ $I = I_R \cosh\sqrt{ZY}s + \frac{E_R}{Z_0}\sinh\sqrt{ZY}s$ This results indicates two solutions, one for the plus sign and the other for the minus sign before the radical.	
2	**From the fundamentals of transmission line theory, derive the expression for input impedance of transmission line.** The input impedance of the Transmission Line is defined as the impedance measured across the input terminals of the transmission line It is denoted as Z in or Z s. It is the ratio of the voltage to the current at the sending end of the transmission line Therefore $Z_{in} = Z_s = \frac{V_s}{I_s}\Omega$ Consider a transmission line of length ‘ with load impedance Z_R. Let V_R and I_R be the receiving end voltage and currents V_R is the voltage across the Z_R, I_R is the current through Z_R  If the transmission line equations are considered interms of receiving end voltage and current by taking s = l We get V = Vs and I = Is	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	$V_s = V_R \cosh(\gamma l) + I_R Z_0 \sinh(\gamma l)$ ----(1) $I_s = I_R \cosh(\gamma l) + \frac{V_R}{Z_0} \sinh(\gamma l)$ ----(2) But, we know that $Z_{in} = Z_s = \frac{V_s}{I_s} \Omega$ $Z_{in} = \frac{V_R \cosh(\gamma l) + I_R Z_0 \sinh(\gamma l)}{I_R \cosh(\gamma l) + \frac{V_R}{Z_0} \sinh(\gamma l)}$ Taking coshγl as common term in numerator and denominator, we get, $Z_{in} = \frac{V_R + I_R Z_0 \frac{\sinh(\gamma l)}{\cosh(\gamma l)}}{I_R + \frac{V_R}{Z_0} \frac{\sinh(\gamma l)}{\cosh(\gamma l)}}$ $Z_{in} = \frac{V_R + I_R Z_0 \tanh(\gamma l)}{I_R + \frac{V_R}{Z_0} \tanh(\gamma l)}$ Multiply numerator and denominator, with Z_0 / I_R , we get	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	$$Z_{in} = \frac{\frac{Z_0}{I_R}[V_R + I_R Z_0 \tanh(\gamma l)]}{\frac{Z_0}{I_R}[I_R + \frac{V_R}{Z_0}\tanh(\gamma l)]}$$ $$Z_{in} = \frac{\frac{Z_0}{I_R}V_R + Z_0^2 \tanh(\gamma l)}{Z_0 + \frac{V_R}{I_R}\tanh(\gamma l)}$$ $$Z_{in} = \frac{Z_0[\frac{V_R}{I_R} + Z_0 \tanh(\gamma l)]}{[Z_0 + \frac{V_R}{I_R}\tanh(\gamma l)]}$$ $$Z_{in} = \frac{Z_0[Z_R + Z_0 \tanh(\gamma l)]}{[Z_0 + Z_R \tanh(\gamma l)]} \quad \because \frac{V_R}{I_R} = Z_R$$ So finally, Input Impedance is expressed as $$Z_{in} = Z_0[\frac{Z_R + Z_0 \tanh(\gamma l)}{Z_0 + Z_R \tanh(\gamma l)}]\Omega$$	
3	**Derive the expression for characteristic impedance of a transmission line.**	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Consider a short line terminated in its characteristic impedance Z_0 as shown in the Fig. 11.5 (a). The short line is a symmetrical network and hence can be represented by the equivalent T-section as shown in the Fig. 11.5 (b). Such a representation is the property of a symmetrical network.	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	 (a) Short line (b) Equivalent T section Fig. 11.5 It is known that finite line terminated in Z_0 behaves as an infinite line hence the input impedance Z_{in} of the equivalent T-section network also must be Z_0. The input impedance Z_{in} of the equivalent T section network can be obtained as, $Z_{in} = \frac{Z_1}{2} + \left\{Z_2 \parallel \left(\frac{Z_1}{2} + Z_0\right)\right\}$ $\therefore \quad Z_{in} = \frac{Z_1}{2} + \frac{Z_2\left(\frac{Z_1}{2} + Z_0\right)}{Z_2 + \frac{Z_1}{2} + Z_0}$ But $\quad Z_{in} = Z_0$ $\therefore \quad Z_0 = \frac{Z_1}{2} + \frac{Z_2\left(\frac{Z_1}{2} + Z_0\right)}{Z_2 + \frac{Z_1}{2} + Z_0}$ $2Z_0\left[Z_2 + \frac{Z_1}{2} + Z_0\right] = Z_1\left(Z_2 + \frac{Z_1}{2} + Z_0\right) + 2Z_2\left(\frac{Z_1}{2} + Z_0\right)$ $\therefore 2Z_0 Z_2 + Z_0 Z_1 + 2Z_0^2 = Z_1 Z_2 + \frac{Z_1^2}{2} + Z_0 Z_1 + Z_1 Z_2 + 2Z_0 Z_2$ $\therefore \quad 2Z_0^2 = 2Z_1 Z_2 + \frac{Z_1^2}{2}$ $\therefore \quad Z_0^2 = \frac{Z_1^2}{4} + Z_1 Z_2 \quad \text{... (1)}$ This is the result applicable for equivalent symmetrical T circuit. Hence Z_0 for the equivalent T section of the finite line is, $Z_0 = \sqrt{\frac{Z_1^2}{4} + Z_1 Z_2} \quad \text{... (2)}$	
4	**Derive the equations of attenuation constant and phase constant of a transmission line in terms of R, L, C & G.** The propagation constant is	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	$\gamma = \alpha + j\beta$ $= \sqrt{(R + j\omega L)(G + j\omega C)}$ $\alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$ $\alpha + j\beta = \sqrt{(RG - \omega^2 LC) + j\omega(LG + RC)}$ Squaring both sides $(\alpha + j\beta)^2 = (RG - \omega^2 LC) + j\omega(LG + RC)$ $\alpha^2 - \beta^2 + 2j\alpha\beta = (RG - \omega^2 LC) + j\omega(LG + RC)$ $\alpha^2 - \beta^2 + 2j\alpha\beta = (RG - \omega^2 LC) + j\omega(LG + RC)$ Equating real terms $\alpha^2 - \beta^2 = RG - \omega^2 LC \quad .....(1)$ Take $\lvert\gamma\rvert = \lvert\alpha + j\beta\rvert = \sqrt{\alpha^2 + \beta^2}$ $\sqrt{\alpha^2 + \beta^2} = \left\lvert\sqrt{(R + j\omega L)(G + j\omega C)}\right\rvert$ Square both sides $\alpha^2 + \beta^2 = \lvert(R + j\omega L)(G + j\omega C)\rvert$	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Now $\alpha^2+\beta^2=\sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}$ .....(2) Add Eq.,(1) &(2) $\alpha^2-\beta^2+\alpha^2+\beta^2=RG-\omega^2LC+\sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}$ $2\alpha^2=RG-\omega^2LC+\sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}$ $2\alpha^2=RG-\omega^2LC+\sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}$ $\alpha^2=\frac{1}{2}[RG-\omega^2LC+\sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}]$ $\alpha=\sqrt{\frac{1}{2}[RG-\omega^2LC+\sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}]}$ Now subtract Eq.,(1) &(2) $\alpha^2-\beta^2-\alpha^2-\beta^2=RG-\omega^2LC-\sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}$ $-2\beta^2=RG-\omega^2LC-\sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}$ $2\beta^2=\omega^2LC-RG+\sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}$	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	$2\beta^2 = \omega^2 LC - RG + \sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}$ $\beta^2 = \frac{1}{2}[\omega^2 LC - RG + \sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}]$ $\beta = \sqrt{\frac{1}{2}[\omega^2 LC - RG + \sqrt{(R^2+\omega^2L^2)(G^2+\omega^2C^2)}]}$	
5	**A lossless transmission line of length 100m has an L of 28μH and a C of 20nF. Find propagation velocity, phase constant and characteristic impedance at an operating frequency of 100KHz.** **Step 1** Calculate the propagation velocity Use the formula: $v_p = \frac{1}{\sqrt{LC}}$ Substitute the given values: $v_p = \frac{1}{\sqrt{(28\times10^{-6}\text{ H/m})(20\times10^{-9}\text{ F/m})}}$ Calculate the result: $v_p = 422577.83$ m/s **Step 2** Calculate the angular frequency Use the formula: $\omega = 2\pi f$ Substitute the given value: $\omega = 2\pi(100\times10^3\text{ Hz})$ Calculate the result: $\omega = 2\pi\times10^5$ rad/s	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Step 3 Calculate the phase constant Use the formula: $\beta = \omega\sqrt{LC}$ Substitute the calculated and given values: $\beta = (2\pi \times 10^5 \text{ rad/s})\sqrt{(28 \times 10^{-6} \text{ H/m})(20 \times 10^{-9} \text{ F/m})}$ Calculate the result: $\beta = 1.485 \text{ rad/m}$ **Step 4** Calculate the characteristic impedance Use the formula: $Z_0 = \sqrt{\frac{L}{C}}$ Substitute the given values: $Z_0 = \sqrt{\frac{28 \times 10^{-6} \text{ H/m}}{20 \times 10^{-9} \text{ F/m}}}$ Calculate the result: $Z_0 = 37.42\ \Omega$ **Solution** The propagation velocity is 422577.83 m/s, the phase constant is 1.485 rad/m, and the characteristic impedance is 37.42 Ω.	
6	**Find the characteristic impedance and propagation constant of a line at 1600Hz, if the following measurements have been made on the at 1600 Hz, Z_{oc} = 650Ω and Z_{sc} = 550Ω.** Given data Frequency (f) = 1600Hz Z_{OC}=650Ω Z_{SC}=550Ω The characteristic impedance of a line s	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	$Z_0 = \sqrt{Z_{0C} \cdot Z_{SC}}$: $Z_0 = \sqrt{650 X 550}$ z = 598 Ω Substitute the values of Z_{OC}, Z_{sc} $\tanh \gamma = \sqrt{\frac{Z_{SC}}{Z_{OC}}}$	
7	**Derive an expression for input impedance, Voltage and current equations of open circuited and shorted circuited transmission line.** Substitute in Tx line equations $Z_R = 0$ FOR SHORT CIRCUIT Substitute in Tx line equations $Z_R = \infty$ FOR OPEN CIRCUIT $V_s = V_R \cosh(\gamma l) + I_R Z_0 \sinh(\gamma l)$ ----(1) $I_s = I_R \cosh(\gamma l) + \frac{V_R}{Z_0} \sinh(\gamma l)$ ----(2)	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	OPEN & SHORT CIRCUITED LINES: INPUT IMPEDANCE $Z_S = Z_O\left[\frac{Z_R \cosh \upsilon l + Z_O \sinh \upsilon l}{Z_O \cosh \upsilon l + Z_R \sinh \upsilon l}\right]$ i) For short circuit line, $Z_R = 0$ $Z_{SC} = Z_O\left[\frac{Z_O \sinh \upsilon l}{Z_O \cosh \upsilon l}\right] = Z_O \tanh \upsilon l$ $Z_{SC} = Z_O \tanh \upsilon l$ ii) For open circuit line, $Z_R = \infty$ $Z_{OC} = \frac{Z_O \cdot Z_R}{Z_R}\left[\frac{\cosh \upsilon l + Z_O/Z_R \sinh \upsilon l}{\frac{Z_O}{Z_R}\cosh \upsilon l + \sinh \upsilon l}\right]$ $Z_{OC} = Z_O \coth \upsilon l$ $Z_{OC} \cdot Z_{SC} = (Z_O \coth \upsilon l)(Z_O \tanh \upsilon l) = Z_O^2$ $Z_O = \sqrt{Z_{OC} \cdot Z_{SC}}$ $\frac{Z_{SC}}{Z_{OC}} = \frac{Z_O \tanh \upsilon l}{Z_O \coth \upsilon l}$ $(\tanh \upsilon l)^2 = \frac{Z_{SC}}{Z_{OC}}$ $\upsilon l = \tanh^{-1}\sqrt{\frac{Z_{SC}}{Z_{OC}}}$	
8	**At 8 MHz the characteristic impedance of transmission line is (40-j2) Ω and the propagation constant is (0.01+j0.18) per meter. Find the primary constants.**	10

MSCE -

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	The primary constants of a transmission line are: • **R**: Resistance per unit length (Ω/m) • **L**: Inductance per unit length (H/m) • **G**: Conductance per unit length (S/m) • **C**: Capacitance per unit length (F/m) We are given: • Characteristic impedance: $$Z_0 = (40 - j2)\ \Omega$$ • Propagation constant: $$\gamma = (0.01 + j0.18) \text{ per meter}$$	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	• Frequency: $$f = 8\ \text{MHz} = 8 \times 10^6\ \text{Hz}$$ • Angular frequency: $$\omega = 2\pi f = 2\pi(8 \times 10^6) = 1.6 \times 10^7\ \text{rad/s}$$ The propagation constant and characteristic impedance are related to the primary constants by: $$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$ $$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$ **Step 1: Compute $R + j\omega L$** $$R + j\omega L = Z_0 \cdot \gamma$$ $$(R + j\omega L) = (40 - j2) \times (0.01 + j0.18)$$ Expanding: $$= (40 \times 0.01 + 40 \times j0.18 - j2 \times 0.01 - j2 \times j0.18)$$ $$= (0.4 + j7.2 - j0.02 + 0.36)$$ $$= (0.76 + j7.18)$$ So, $$R = 0.76\ \Omega/m, \quad \omega L = 7.18$$ $$L = \frac{7.18}{\omega} = \frac{7.18}{1.6 \times 10^7} = 4.49 \times 10^{-7}\ \text{H/m}$$ **Step 2: Compute $G + j\omega C$**	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	$$G + j\omega C = \frac{\gamma}{Z_0}$$ $$G + j\omega C = \frac{(0.01 + j0.18)}{(40 - j2)}$$ Multiplying numerator and denominator by the complex conjugate of the denominator: $$= \frac{(0.01 + j0.18)(40 + j2)}{(40 - j2)(40 + j2)}$$ $$= \frac{0.01 \times 40 + 0.01 \times j2 + j0.18 \times 40 + j0.18 \times j2}{1600 + 4}$$ $$= \frac{0.4 + j0.02 + j7.2 - 0.36}{1604}$$ $$= \frac{0.04 + j7.22}{1604}$$ $$= (2.49 \times 10^{-5} + j4.5 \times 10^{-3})$$ So, $$G = 2.49 \times 10^{-5}\ \text{S/m}, \quad \omega C = 4.5 \times 10^{-3}$$ $$C = \frac{4.5 \times 10^{-3}}{\omega} = \frac{4.5 \times 10^{-3}}{1.6 \times 10^{7}} = 2.81 \times 10^{-10}\ \text{F/m}$$ **Final Answer:** $$R = 0.76\ \Omega/m, \quad L = 449\ \text{nH/m}, \quad G = 24.9\ \mu\text{S/m}, \quad C = 281\ \text{pF/m}$$	
9	**A 30 m long lossless line with ZO = 50 Ω operating at 2 MHz is terminated with a load Z_L = 60+j40 Ω. If v = 0.6C on the line, find: (i) Reflection coefficient. (ii) Standing wave ratio. (iii) Input Impedance.** **Ans:**Given data A lossless transmission line length (l) = 30m The Characteristic Impedance is	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	$Z_O = 50\ \Omega$ Terminated load is $Z_L = 60+j40\ \Omega$ Operating frequency (f) = 2MHz Wave Velocity (v) is v = 0.6C = 0.6 X $3X10^8$ m/sec For a lossless transmission line, $\gamma = \alpha + j\beta = j\beta \quad (\because \alpha = 0)$ $\gamma l = j\beta l$ i) Reflection coefficient (Γ): $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{60 + j40 - 50}{60 + j40 + 50} = \frac{10 + j40}{110 + j40} = 0.3523\angle 56^0$ ii) Standing wave ratio (s): $s = \frac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert} = \frac{1 + 0.3523}{1 - 0.3523} = 2.088$ iii) Input Impedance: $\beta l = \frac{\omega}{u} l = \frac{2\pi(2\times 10^6)}{0.6(3\times 10^8)}(30) = \frac{2\pi}{3} = 120^0$ (electrical length) $Z_{in} = Z_0\left[\frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}\right]$ $= 50\left[\frac{60 + j40 + j50\tan 120^0}{50 + j(60 + j40)\tan 120^0}\right]$ $= 23.97 + j1.35 = 24.01\angle 3.22^0$	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
10	**Describe the losses in transmission lines.** Losses in transmission lines are of three types: Losses Copper losses — Dielectric losses — Radiation losses **Copper Loss** These losses occur because of the following reasons: 1. I^2R **power loss** This is due to dissipation as a result of heating in pure resistance. Its features are: (a) Copper loss is small if there are no current loops, that is, the line should be properly terminated without producing standing waves. (b) Copper loss is more if z_0 is small. This is because, if z_0 is low, current is high. If the current is high, copper loss is more. 2. **Skin effect** Its features are: (a) When an AC signal at high frequency is applied to the transmission lines, the current is confined to the surface (skin) of the conductors. This is known as skin effect. This effect reduces the cross-sectional area of the conductor with increase in frequency. (b) Decrease in cross-sectional area increases the resistance. (c) Increased resistance increases power losses.	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	3. **Crystallisation** Its features are: (a) Copper losses increase due to ageing of the transmission line. Losses are more when the line is subjected to high temperature, high winds and moisture. Moreover, bending of the line back and forth causes the line to become brittle and cracks appear. This effect is known as crystallisation of the conductors. (b) Crystallisation increases resistance in the conductors which in turn increases copper losses. **Dielectric Losses** These losses exist due to improper characteristics of dielectric. Salient features: (a) These are due to $I^2 R$ power dissipation because of the heating of the solid dielectric material between conductors in transmission lines. These losses are proportional to the voltage across the dielectric. (b) With increased frequencies, solid dielectric properties worsen and hence transmission lines with solid dielectrics have limited applications. (c) Lines with air dielectric are used at high frequencies, as air dielectric loss is very small. **Radiation Losses** Salient features: (a) These losses are high when the spacing between the lines is high as the transmission line acts as an antenna. Therefore, radiation losses are more in parallel-wire lines than in coaxial lines. (b) At high frequency, λ will be small and hence the transmission lines are not useful at high frequencies.	
11	**A Transmission line operating at 500 MHz has Z_0 = 80 Ω, α = 0.04 Nepers/mand β = 1.5rad/m then find line parameters?**	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	The primary parameters of a transmission line are: • **R**: Resistance per unit length (Ω/m) • **L**: Inductance per unit length (H/m) • **G**: Conductance per unit length (S/m) • **C**: Capacitance per unit length (F/m) **Given Data:** • Characteristic impedance: $Z_0 = 80\ \Omega$ • Propagation constant: $\gamma = \alpha + j\beta = 0.04 + j1.5 \text{ per meter}$ • Frequency: $f = 500 \text{ MHz} = 500 \times 10^6 \text{ Hz}$ • Angular frequency: $\omega = 2\pi f = 2\pi(500 \times 10^6) = 3.14 \times 10^9 \text{ rad/s}$	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	The relationships between primary and secondary constants are: $$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$ $$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$ **Step 1: Compute $R + j\omega L$** $$R + j\omega L = Z_0 \cdot \gamma$$ $$(R + j\omega L) = (80) \times (0.04 + j1.5)$$ $$= 3.2 + j120$$ So, $$R = 3.2\ \Omega/m, \quad \omega L = 120$$ $$L = \frac{120}{\omega} = \frac{120}{3.14 \times 10^9} = 38 \times 10^{-9}\ \mathrm{H/m} = 38.2\ \mathrm{nH/m}$$	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Step 2: Compute $G + j\omega C$ $$G + j\omega C = \frac{\gamma}{Z_0}$$ $$G + j\omega C = \frac{(0.04 + j1.5)}{80}$$ $$= (5 \times 10^{-4} + j0.01875)$$ So, $$G = 5 \times 10^{-4}\ \mathrm{S/m}, \quad \omega C = 0.01875$$ $$C = \frac{0.01875}{\omega} = \frac{0.01875}{3.14 \times 10^{9}} = 5.97 \times 10^{-12}\ \mathrm{F/m} = 5.97\ \mathrm{pF/m}$$ Final Answer: $$R = 3.2\ \Omega/m, \quad L = 38.2\ \mathrm{nH/m}, \downarrow = 0.5\ \mathrm{mS/m}, \quad C = 5.97\ \mathrm{pF/m}$$	
12	**What is meant by distortion less transmission line and derive the expression for condition for it.** A line, which has neither frequency distortion nor phase distortion is called a distortion less line. The condition for a distortion less line is RC=LG. Also, a) The attenuation constant should be made independent of frequency. b) The phase constant should be made dependent of frequency. c) The velocity of propagation is independent of frequency.	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	CONDITION FOR DISTORTIONLESS LINE: W.K.T $\beta = \sqrt{\dfrac{(w^2LC - RG) + \sqrt{(RG - w^2LC)^2 + w^2(LG + RC)^2}}{2}}$ 1 To bring β, a direct function of frequency, the term under the second radical be reduced to $(RG + w^2LC)^2$ $(RG - w^2LC)^2 + w^2(LG + RC)^2 = (RG + w^2LC)^2$ 2 $R^2G^2 + w^4L^2C^2 - 2w^2RG\,LC + w^2L^2G^2 + w^2R^2C^2 + 2w^2LG\,RC = R^2G^2 + w^4L^2C^2 + 2w^2RG\,LC$ $w^2L^2G^2 + w^2R^2C^2 - 2w^2RG\,LC = 0$ $L^2G^2 + R^2C^2 - 2\,RG\,LC = 0$ $(LG - RC)^2 = 0$ $LG = RC$ 3 ∴ the condition that makes β, a direct function of frequency is LG = RC Sub eq. 2 in 1 $\beta = \sqrt{\dfrac{(w^2LC - RG) + \sqrt{(RG + w^2LC)^2}}{2}}$ $\beta = \sqrt{\dfrac{(w^2LC - RG) + \sqrt{(RG + w^2LC)}}{2}} = \sqrt{\dfrac{2w^2LC}{2}}$ $\beta = w\sqrt{LC}$ 4	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Velocity of propagation $\vartheta = \frac{w}{\beta}$ $\vartheta = \frac{w}{w\sqrt{LC}}$ $\vartheta = \frac{1}{\sqrt{LC}}$ 5 Velocity of propagation (ϑ) is the same for all frequency. Thus eliminating delay distortion. To make α independent of frequency, $\alpha = \sqrt{\frac{(RG - w^2LC)^2 + \sqrt{(RG - w^2LC)^2 + w^2(LG + RC)^2}}{2}}$ The term under the second radical forced to be equal to $(RG + w^2LC)^2$ $\alpha = \sqrt{\frac{(RG - w^2LC) + \sqrt{(RG + w^2LC)^2}}{2}}$ $\alpha = \sqrt{\frac{(RG - w^2LC) + (RG + w^2LC)}{2}} = \sqrt{\frac{2RG}{2}}$ $\alpha = \sqrt{RG}$ 6 From 3 LG = RC $\frac{L}{C} = \frac{R}{G}$ 7 To achieve eq. 7 requires large L since G is small. If G intentionally increased, α is increased results in poor line efficiency. Reduction in R raises the size and thus the cost of conductors, so that above condition poses practical problems.	
1	**Determine the reflection coefficient and voltage standing wave ratio when:** **(i) Z_L = short circuit.** **(ii) Z_L = open circuit.** **(iii) Z_L = purely reactive.**	10

CE - 1101

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	(iv) $Z_L = Z_0$. The voltage reflection coefficient Γ is $$\Gamma = \frac{V_0^-}{V_0^+} \equiv \left(\frac{Z_L - Z_0}{Z_L + Z_0}\right) = \frac{z_L - 1}{z_L + 1},$$ z_L is Z_L/Z_0 is the normalized load impedance $$z_L = Z_L/Z_0 = (R + jX)/Z_0 = r + jx$$ i) Z_L = short circuit $Z_L = 0$ $$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{0 - Z_0}{0 + Z_0} = -1$$ Polar form Γ can be written as $\Gamma = \lvert\Gamma\rvert\angle\theta$ $$\Gamma = \lvert\Gamma\rvert\angle\theta = 1\angle 180^0$$ ii) Z_L = open circuit. $Z_L = \infty$ $$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{\infty - Z_0}{\infty + Z_0} = 1$$ Polar form Γ can be written as $\Gamma = \lvert\Gamma\rvert\angle\theta$ $$\Gamma = \lvert\Gamma\rvert\angle\theta = 1\angle 0^0$$ iii) Z_L = purely reactive. $Z_L = jX = j\omega L$ (or) $Z_L = jX = -j/\omega C$ $$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{jX - Z_0}{jX + Z_0}$$	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q

As we know

$$|\Gamma|^2 = \Gamma\Gamma^* = \frac{(jX - Z_0)}{(jX + Z_0)}\frac{(-jX - Z_0)}{(-jX + Z_0)} = \frac{X^2 - jXZ_0 + jXZ_0 + Z_0^{\ 2}}{X^2 + Z_0^{\ 2}} = 1$$

$$|\Gamma| = 1$$

iv) $Z_L = Z_O$. (Matched Line)

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{Z_0 - Z_0}{Z_0 + Z_0} = 0$$

Reflection Coefficient $\Gamma = |\Gamma|e^{j\theta_r}$

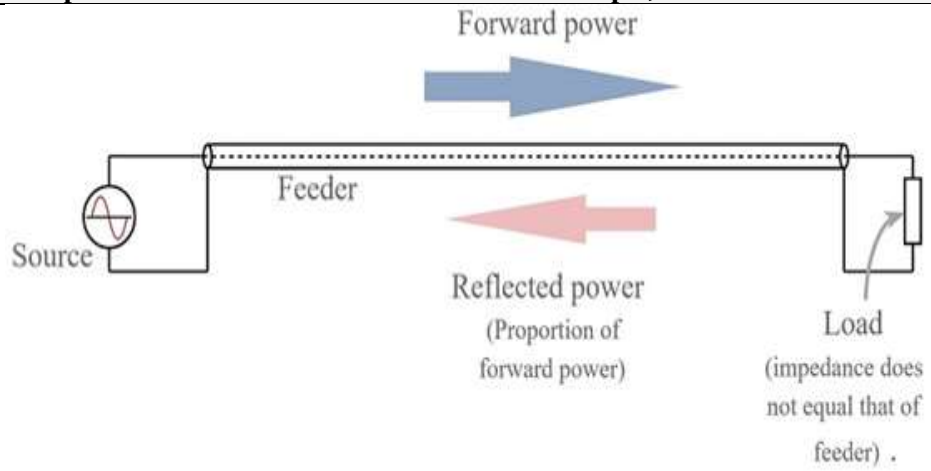
Load	$\lvert\Gamma\rvert$	θ_r
Z_0 $Z_L = (r + jx)Z_0$	$\left[\frac{(r-1)^2 + x^2}{(r+1)^2 + x^2}\right]^{1/2}$	$\tan^{-1}\left(\frac{x}{r-1}\right) - \tan^{-1}\left(\frac{x}{r+1}\right)$
Z_0 Z_0	0 (no reflection)	irrelevant
Z_0 (short)	1	$\pm 180°$ (phase opposition)
Z_0 (open)	1	0 (in-phase)
Z_0 $jX = j\omega L$	1	$\pm 180° - 2\tan^{-1} x$
Z_0 $jX = \frac{-j}{\omega C}$	1	$\pm 180° + 2\tan^{-1} x$

The voltage standing wave ratio (VSWR) or (SWR) is

$$S = \frac{|\tilde{V}|_{\max}}{|\tilde{V}|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	i) Z_L = short circuit $$\lvert\Gamma\rvert = 1$$ $$S = \frac{1+\lvert\Gamma\rvert}{1-\lvert\Gamma\rvert} = \frac{1+1}{1-1} = \infty$$ ii) Z_L = open circuit. $$\lvert\Gamma\rvert = 1$$ $$S = \frac{1+\lvert\Gamma\rvert}{1-\lvert\Gamma\rvert} = \frac{1+1}{1-1} = \infty$$ iii) Z_L = purely reactive. $$\lvert\Gamma\rvert = 1$$ $$S = \frac{1+\lvert\Gamma\rvert}{1-\lvert\Gamma\rvert} = \frac{1+1}{1-1} = \infty$$ iv) $Z_L = Z_O$. (Matched Line) $$\lvert\Gamma\rvert = 0$$ $$S = \frac{1+\lvert\Gamma\rvert}{1-\lvert\Gamma\rvert} = \frac{1+0}{1-0} = 1$$	
2	**a) Explain why short circuited stubs are preferred over open circuited stub.** In transmission lines, short-circuited stubs are preferred over open-circuited stubs due to several practical and technical reasons: **1. Lower Radiation Losses** Open-circuited stubs have an exposed end, which can radiate electromagnetic energy, leading to signal loss and unwanted interference.	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Short-circuited stubs, on the other hand, have a solid electrical connection at the termination, preventing radiation losses. **2. Better High-Frequency Performance** At high frequencies, open-circuited stubs exhibit parasitic capacitance at the open end, which alters their impedance characteristics. Short-circuited stubs do not suffer from these parasitic effects, making their impedance more predictable and stable. **3. Ease of Fabrication and Implementation** In practical transmission line systems, implementing a short-circuit by connecting to a ground plane is simpler and more reliable than achieving a perfect open circuit. Achieving a true open circuit is difficult because of the presence of stray capacitance at the open end. **4. Reduced Sensitivity to Environmental Factors** Open-circuited stubs are more affected by environmental changes such as temperature, humidity, and nearby conductors due to their exposed nature. Short-circuited stubs are less sensitive because they are grounded, making them more stable in varying environmental conditions. **5. Elimination of Parasitic Effects** Open-circuited stubs introduce fringing fields and parasitic capacitance at the open end, affecting the expected impedance values. Short-circuited stubs avoid this issue, leading to better impedance matching and predictable performance. **6. Reduced Noise and Interference Pickup** Open-ended stubs can act as antennas, picking up unwanted external signals and noise. Short-circuited stubs, being connected to ground, have better noise immunity. **Conclusion:** In transmission lines, short-circuited stubs are preferred because they exhibit lower radiation losses, better high-frequency stability, ease of implementation, and reduced environmental sensitivity. These advantages make them more reliable for impedance matching, filter design, and transmission line tuning applications. **b) Describe all the characteristics of UHF Lines**	
3	**What is the significance of standing wave ratio in a transmission line.** **Calculate the reflection coefficient and VSWR for a 50Ω line terminated with i) Matched load ii) short circuit iii) +j75Ω load iv) –j75Ω load** **Voltage Standing Wave Ratio (VSWR)** is defined as the ratio between the incident and reflected wave in an RF component or system. It determines the efficiency of power transmission from a source to the load through a transmission line (or cable). The minimum value of VSWR can be 1:1 (or simply 1), which is an ideal condition where 100% of power from source is absorbed by the load. However, in real-world applications, VSWR is rarely found to be 1:1 and systems are designed to keep the VSWR as close to unity as possible.	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	 In an ideal RF transmission system, the impedance of source and transmission line (or cable) must be perfectly matched with the impedance of the connected load. VSWR can be defined as the numerical measure of how well the impedances of these integral components are matched. **VSWR Calculation** There is a mathematical equation to accurately calculate the VSWR of an RF system. $$\text{VSWR} = \frac{1+\lvert\Gamma\rvert}{1-\lvert\Gamma\rvert}$$; where Γ is the reflection coefficient. Let's calculate the reflection coefficient (Γ) and Voltage Standing Wave Ratio (VSWR) for a 50Ω transmission line terminated with different loads.	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Formulae Used 1. **Reflection Coefficient (Γ):** $$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$ where: • Z_L = Load Impedance • $Z_0 = 50\Omega$ (Characteristic Impedance) 2. **Voltage Standing Wave Ratio (VSWR):** $$\text{VSWR} = \frac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert}$$	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Case (i): Matched Load ($Z_L = 50\Omega$) $$\Gamma = \frac{50-50}{50+50} = \frac{0}{100} = 0$$ $$\text{VSWR} = \frac{1+0}{1-0} = 1$$ ✅ **Result:** $\Gamma = 0$, **VSWR = 1** (Perfect Match, No Reflections) **Case (ii): Short Circuit ($Z_L = 0$)** $$\Gamma = \frac{0-50}{0+50} = \frac{-50}{50} = -1$$ $$\text{VSWR} = \frac{1+1}{1-1} = \infty$$ ✅ **Result:** $\Gamma = -1$, **VSWR = ∞** (Complete Reflection)	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Case (iii): Load $Z_L = +j75\Omega$ $$\Gamma = \frac{j75 - 50}{j75 + 50}$$ Multiplying numerator and denominator by the complex conjugate $j75 - 50$: $$\Gamma = \frac{(j75 - 50)(j75 - 50)}{(j75 + 50)(j75 - 50)}$$ $$= \frac{j^2 5625 - 2j(75)(50) + 2500}{j^2 5625 - 2500}$$ $$= \frac{-5625 - j7500 + 2500}{-5625 - 2500}$$ $$= \frac{-3125 - j7500}{-8125}$$ $$= 0.384 - j0.923$$ $$\lvert\Gamma\rvert = \sqrt{(0.384)^2 + (-0.923)^2} = \sqrt{0.147 + 0.852} = \sqrt{0.999} \approx 1$$ $$\text{VSWR} = \frac{1+1}{1-1} = \infty$$ ✅ **Result:** $\Gamma \approx 1\angle -67.4^\circ$, **VSWR** = ∞ (Resonant Condition, High Mismatch)	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Case (iv): Load $Z_L = -j75\Omega$ $$\Gamma = \frac{-j75 - 50}{-j75 + 50}$$ Following similar steps: $$\Gamma = \frac{-j75 - 50}{-j75 + 50}$$ Multiplying numerator and denominator by $j75 + 50$: $$\Gamma = \frac{(-j75 - 50)(-j75 + 50)}{(-j75 + 50)(-j75 - 50)}$$ $$= \frac{j^2 5625 + 2j(75)(50) + 2500}{j^2 5625 - 2500}$$ $$= \frac{-5625 + j7500 + 2500}{-8125}$$ $$= \frac{-3125 + j7500}{-8125}$$ $$= 0.384 + j0.923$$ $$\lvert\Gamma\rvert = \sqrt{(0.384)^2 + (0.923)^2} = \sqrt{0.999} \approx 1$$ $$\text{VSWR} = \frac{1+1}{1-1} = \infty$$	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	✅ **Result:** $\Gamma \approx 1\angle 67.4^\circ$, **VSWR** = ∞ (Resonant Condition, High Mismatch) **Final Answers:** Case | Load Z_L | Reflection Coefficient Γ | VSWR (i) Matched Load | 50Ω | 0 | 1 (ii) Short Circuit | 0Ω | -1 | ∞ (iii) $+j75\Omega$ | $1\angle -67.4^\circ$ | ∞ (iv) $-j75\Omega$ | $1\angle 67.4^\circ$ | ∞	
4	A low loss transmission line of 100 Ω characteristic impedance is connected to a load of 400 Ω. Calculate the reflection coefficient and standing wave ratio. Derive the Relationships used.	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Given Data: • Characteristic Impedance: $Z_0 = 100\Omega$ • Load Impedance: $Z_L = 400\Omega$ --- **Step 1: Derive the Reflection Coefficient (Γ) Formula** The **reflection coefficient** (Γ) is defined as: $$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$ where: • Z_L = Load impedance • Z_0 = Characteristic impedance **Substituting the given values:** $$\Gamma = \frac{400 - 100}{400 + 100}$$ $$\Gamma = \frac{300}{500} = 0.6$$ ✅ Reflection Coefficient: $$\Gamma = 0.6$$	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Step 2: Derive the Voltage Standing Wave Ratio (VSWR) Formula The VSWR is related to the reflection coefficient as: $$\text{VSWR} = \frac{1+\lvert\Gamma\rvert}{1-\lvert\Gamma\rvert}$$ **Substituting $\Gamma = 0.6$:** $$\text{VSWR} = \frac{1+0.6}{1-0.6} = \frac{1.6}{0.4} = 4$$ ✅ VSWR: $$\text{VSWR} = 4$$ **Final Answers:** • **Reflection Coefficient,** $\Gamma = 0.6$ • **Voltage Standing Wave Ratio,** VSWR=4	
5	**Explain the significance and utility of λ/8, λ/4 and λ/2 lines.** Transmission lines are used to transmit electromagnetic waves. λ/8, λ/4, and λ/2 represent specific lengths of transmission lines that exhibit unique impedance characteristics, impacting signal reflection and transmission. These lengths are crucial in designing matching networks and impedance transformers. In RF and microwave engineering, transmission line sections of specific electrical lengths (λ/8, λ/4, and λ/2) have unique impedance transformation properties. These properties are widely used in impedance matching, filters, and antenna design.	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Step 1: Identify the Transmission Line Lengths Three transmission line lengths are given: λ/8, λ/4, and λ/2, where λ represents the wavelength at a specific frequency. ***Step 2: Define the Function of λ/8 Line*** A λ/8 transmission line is used for impedance transformation and matching loads to transmission lines. • **Electrical Length:** 45^0 phase shift • **Function:** Provides a **mild impedance transformation** but does not completely invert the impedance. • **Application:** Used in **impedance tapering** or **progressive impedance matching** to gradually transition impedance without strong reflections. **Impedance Transformation Formula:** $$Z_{\text{in}} = Z_0 \frac{Z_L + jZ_0 \tan(\pi/4)}{Z_0 + jZ_L \tan(\pi/4)}$$ $$Z_{\text{in}} = Z_0 \frac{Z_L + jZ_0}{Z_0 + jZ_L}$$ • The impedance transformation is intermediate between the input and load impedances. • Often used in **broadband matching networks**. ***Step 3: Define the Function of λ/4 Line***	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	A λ/4 transmission line is used for impedance matching, impedance transformation, and creating phase shifts. • **Electrical Length:** 90^0 phase shift • **Function: Impedance inversion** – transforms a high impedance to low and vice versa. • **Application:** Used for **impedance matching**, **stub tuning**, and designing **antenna feed networks**. **Impedance Transformation Formula:** $$Z_{in} = \frac{Z_0^2}{Z_L}$$ **Key Properties:** • A **short circuit** at the load appears as an **open circuit** at the input. • A **high impedance** is transformed into a **low impedance**. • Used in **antenna matching**, **power dividers**, and **bandpass filters**. **Example Applications:** 1. **Matching a 300Ω antenna to a 75Ω transmission line** • Choose $Z_0 = \sqrt{300 \times 75} = 150\Omega$ as the quarter-wave matching section. 2. **Short-circuited stub tuning** • A $\lambda/4$ short-circuited stub acts as an **open circuit** at the connection point. ***Step 4: Define the Function of λ/2 Line*** A λ/2 transmission line is used for creating phase shifts, impedance transformation, and designing filters.	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	• **Electrical Length:** 180^0 phase shift • **Function: Repeats the load impedance** at the input. • **Application:** Used for **resonators**, **impedance repeating**, and **antenna feed structures**. **Impedance Transformation Formula:** $$Z_{\text{in}} = Z_L$$ **Key Properties:** • The input impedance is **exactly the same** as the load impedance. • A **short circuit remains a short circuit**, and an **open circuit remains an open circuit**. • Used in **resonant circuits**, **antenna feed design**, and **filters**. **Example Applications:** 1. **Dipole Antennas** • A **λ/2 dipole antenna** resonates at the fundamental frequency. 2. **Stub Tuning and Filters** • $\lambda/2$ **resonators** are used in **bandpass filters**. ***Final Answer*** λ/8, λ/4, and λ/2 transmission lines are used for impedance matching, impedance transformation, phase shifting, and filter design in RF and microwave engineering.	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Summary of Utility Line Length | Function | Application $\lambda/8$ | Mild Impedance Transformation | Broadband Matching, Impedance Tapering $\lambda/4$ | Impedance Inversion | Quarter-Wave Transformers, Stub Matching $\lambda/2$ | Impedance Repetition | Resonators, Antennas, Filters	
6	**Define the reflection coefficient and derive the expression for i/p impedance in terms of reflection coefficient.** The ratio of the amplitudes of the reflected and incident voltage waves at the receiving end of the line is called the **reflection coefficient**. It is denoted by K $K = \frac{\text{reflected voltage at load}}{\text{incident voltage at load}}$ The reflected voltage at load is component E_2 at the receiving end i.e. with $s = 0$. $\therefore \quad E_2\|_{s=0} = \frac{E_R (Z_R - Z_0)}{2 Z_R}$ ... (7) The incident voltage at load is component E_1 at the receiving end i.e. with $s = 0$. $\therefore \quad E_1\|_{s=0} = \frac{E_R (Z_R + Z_0)}{2 Z_R}$ ... (8) $\therefore \quad K = \frac{E_R (Z_R - Z_0) / 2 Z_R}{E_R (Z_R + Z_0) / 2 Z_R}$ $\therefore \quad K = \frac{Z_R - Z_0}{Z_R + Z_0}$ ... (9) It is a measure of the mismatch between the load impedance Z_R and the characteristic impedance Z_0 of the line.	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	11.18.4 Input Impedance Interms of Z_0 and K The input impedance at the sending end is given by, $$Z_S = \frac{E_S}{I_S} = \frac{Z_0\left[Z_R \cosh(\gamma l) + Z_0 \sinh(\gamma l)\right]}{\left[Z_0 \cosh(\gamma l) + Z_R \sinh(\gamma l)\right]}$$ $$\therefore \quad Z_S = \frac{Z_0\left[Z_R + Z_0 \tanh(\gamma l)\right]}{\left[Z_0 + Z_R \tanh(\gamma l)\right]}$$ But $\tanh(\gamma l) = \frac{e^{\gamma l} - e^{-\gamma l}}{e^{\gamma l} + e^{-\gamma l}}$ $$\therefore \quad Z_S = \frac{Z_0\left[Z_R + Z_0\left(\frac{e^{\gamma l} - e^{-\gamma l}}{e^{\gamma l} + e^{-\gamma l}}\right)\right]}{\left[Z_0 + Z_R\left(\frac{e^{\gamma l} - e^{-\gamma l}}{e^{\gamma l} + e^{-\gamma l}}\right)\right]}$$ $$= Z_0\left\{\frac{\left(e^{\gamma l} + e^{-\gamma l}\right)Z_R + Z_0\left(e^{\gamma l} - e^{-\gamma l}\right)}{Z_0\left(e^{\gamma l} + e^{-\gamma l}\right) + Z_R\left(e^{\gamma l} - e^{-\gamma l}\right)}\right\}$$	

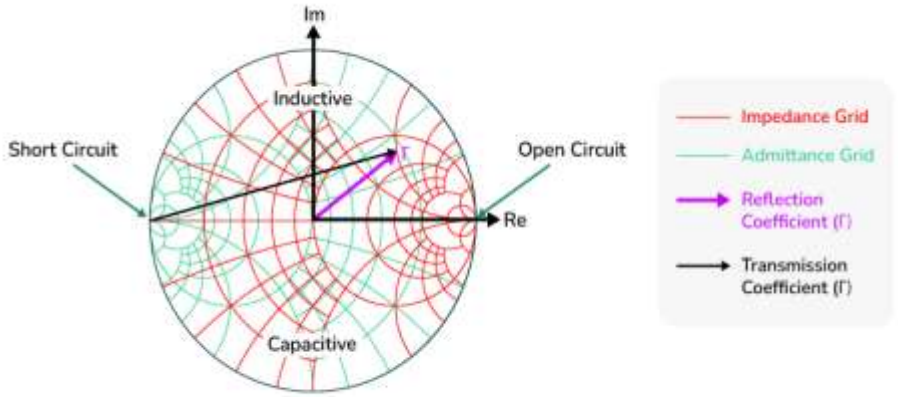
Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	$= Z_0 \left\{ \frac{e^{\gamma l}[Z_R + Z_0] + e^{-\gamma l}[Z_R - Z_0]}{e^{\gamma l}[Z_R + Z_0] + e^{-\gamma l}[Z_0 - Z_R]} \right\}$ $= Z_0 \left\{ \frac{e^{\gamma l} + \frac{Z_R - Z_0}{Z_R + Z_0} e^{-\gamma l}}{e^{\gamma l} - \frac{Z_R - Z_0}{Z_R + Z_0} e^{-\gamma l}} \right\}$ $\therefore \quad Z_S = Z_0 \left\{ \frac{e^{\gamma l} + K e^{-\gamma l}}{e^{\gamma l} - K e^{-\gamma l}} \right\}$	
7	Explain how the i/p impedance varies with the frequency with sketches.	10
8	Explain the significance and design of single stub impedance Matching. Discuss the factors on which stub length depends.	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q

	1. Significance of Single Stub Matching
	In RF and microwave engineering, impedance mismatches cause **signal reflections** and **power loss** in transmission lines. A **single stub tuner** is a simple and effective method to achieve **impedance matching** between a transmission line and a load.
	• **Purpose:** Eliminates reflections by transforming the **load impedance** (Z_L) to match the **characteristic impedance** (Z_0) of the transmission line.
	• **Advantages:**
	  • Simple and adjustable.
	  • Does not require lumped components.
	  • Works efficiently for narrowband applications.
	• **Applications:**
	  • Antenna matching networks.
	  • RF circuits (e.g., power amplifiers).
	  • Microwave systems.
	2. Design of Single Stub Impedance Matching
	A **single stub tuner** consists of:
	1. A **main transmission line** of characteristic impedance Z_0.
	2. A **short-circuited or open-circuited stub** (a transmission line section) placed at a specific distance d from the load.
	Steps for Designing a Single Stub Match
	1. **Determine Load Impedance** (Z_L)
	  • Measure or calculate the complex impedance at the load.
	2. **Calculate Normalized Impedance** (z_L)
	  • Convert Z_L to normalized form using:
	$$z_L = \frac{Z_L}{Z_0}$$	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	3. **Locate the Load Impedance on a Smith Chart** • Plot z_L and move **clockwise** along the **constant** $\lvert\Gamma\rvert$ circle. 4. **Find the Nearest Admittance Point on the Unit Conductance Circle** • Move **toward the generator** along a constant $\lvert\Gamma\rvert$ circle until you reach a **purely susceptive** admittance (jB). 5. **Determine the Distance d from the Load** • The movement distance d corresponds to a physical length on the transmission line: $$d = \frac{\lambda}{2\pi}\theta$$ • where θ is the phase shift along the transmission line. 6. **Calculate the Stub Reactance X_{stub}** • The stub provides the negative reactance needed to cancel out jB: $$X_{\text{stub}} = -jB$$ 7. **Determine the Stub Length l** • Use transmission line equations to find l for a short-circuited or open-circuited stub: • **Short-circuited stub:** $$l = \frac{\lambda}{2\pi}\tan^{-1}\left(\frac{X_{\text{stub}}}{Z_0}\right)$$ • **Open-circuited stub:** $$l = \frac{\lambda}{2\pi}\cot^{-1}\left(\frac{X_{\text{stub}}}{Z_0}\right)$$ 8. **Implement the Stub** • The stub (either short-circuited or open-circuited) is placed at distance d from the load to provide the required reactance.	

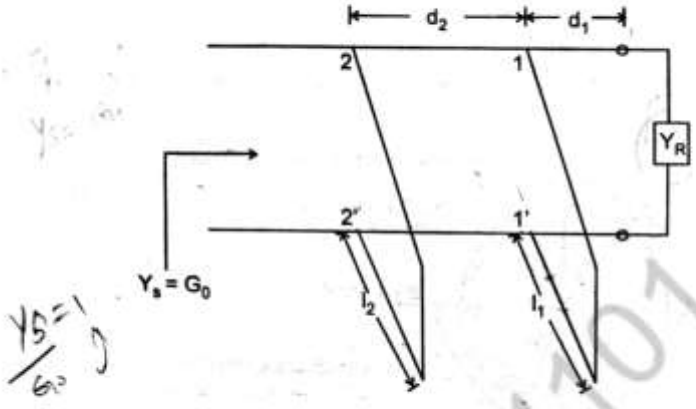
Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	3. Factors Affecting Stub Length The stub length l depends on several factors: 1. **Frequency of Operation (f)** • Since stub lengths are based on **wavelength** λ ($\lambda = \frac{c}{f}$), higher frequencies require shorter stubs. 2. **Characteristic Impedance (Z_0)** • The stub reactance is a function of Z_0, affecting the required stub length. 3. **Load Impedance (Z_L)** • The more mismatched Z_L is, the longer the stub required for compensation. 4. **Stub Type (Short-Circuited vs. Open-Circuited)** • **Short-circuited stubs** are preferred at high frequencies because they reduce radiation losses. • **Open-circuited stubs** are easier to fabricate but may introduce parasitic capacitances. 5. **Physical Constraints** • Practical implementation may require stubs to be shortened using additional components (e.g., lumped elements in PCB circuits).	
9	**Explain smith chart. Write steps to determine VSWR, the input impedance and reflection coefficient.** The Smith Chart is a graphical tool that is used in RF transmission line design and electrical engineering. It helps in analyzing and designing transmission lines and impedance-matching networks. It represents complex impedance values on a polar plot, which allows experts to visualize and manipulate impedance changes.	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	 Smith Chart By using the Smith Chart, engineers can improve signal transmission and minimize reflections, which is particularly important in applications like antenna design, microwave circuits, and radio frequency systems. The Smith Chart simplifies complex calculations involved in designing and analyzing impedance-matching circuits and transmission lines, making it an important tool in RF engineering and microwave design. The **Smith Chart** is a powerful graphical tool used in RF engineering to determine the **reflection coefficient (Γ\GammaΓ)**, **Voltage Standing Wave Ratio (VSWR)**, and **input impedance (ZinZ_{\text{in}}Zin)** of a transmission line.	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Step 1: Normalize the Load Impedance Before plotting on the Smith Chart, **normalize the load impedance** with respect to the characteristic impedance Z_0: $$z_L = \frac{Z_L}{Z_0}$$ where: • Z_L = Load impedance • Z_0 = Characteristic impedance of the transmission line • z_L = Normalized impedance Find this normalized impedance on the Smith Chart. **Step 2: Locate the Load Impedance on the Smith Chart** • Find the **real part** of z_L on the **resistance circles**. • Find the **imaginary part** on the **reactance arcs**. • The intersection gives the point corresponding to z_L. --- **Step 3: Determine the Reflection Coefficient (Γ_L)** • From the plotted load impedance, **draw a line from the center of the Smith Chart to the load impedance point**. • The distance from the center to this point gives **the magnitude** of the reflection coefficient $\lvert\Gamma_L\rvert$. • The angle (measured counterclockwise from the positive real axis) gives **the phase angle** θ of Γ_L. ↓ Thus,	

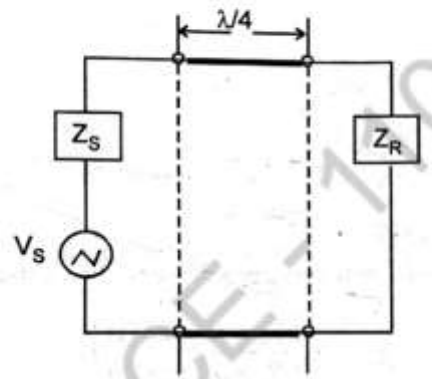
Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Thus, $$\Gamma_L = \lvert\Gamma_L\rvert e^{j\theta}$$ **Step 4: Find VSWR** VSWR is related to the reflection coefficient as: $$\text{VSWR} = \frac{1+\lvert\Gamma_L\rvert}{1-\lvert\Gamma_L\rvert}$$ To find **VSWR** using the Smith Chart: • Measure the **distance of Γ_L from the center**. • Use the **constant VSWR circles** on the chart. • The intersection of this radius with the **horizontal axis** gives the **VSWR value**. **Step 5: Find Input Impedance at a Distance l from Load** • **Move along the constant $\lvert\Gamma\rvert$ circle clockwise** by $2\beta l$ (where $\beta = 2\pi/\lambda$). • Each **full rotation (360°)** corresponds to **half-wavelength ($\lambda/2$)**. • After moving the required distance, read the new impedance z_{in} from the chart. Convert back to actual impedance: $$Z_{\text{in}} = z_{\text{in}} Z_0$$	

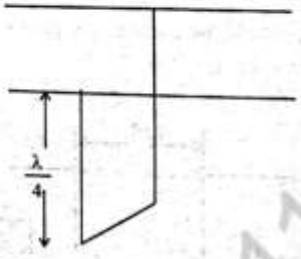
Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Summary of Steps 1. Normalize the load impedance $z_L = Z_L/Z_0$. 2. Plot z_L on the Smith Chart. 3. Determine the reflection coefficient Γ_L by measuring the distance from the center and finding the angle. 4. Find VSWR using the constant VSWR circles. 5. Move along the constant $\lvert\Gamma\rvert$ circle to find the input impedance at any distance l.	
10	Explain how the i/p impedance varies with the frequency with sketches	10
11	**Write a detailed note on double stub matching on a line using smith chart.** **Limitations of Single Stub Matching** i) The stub has to be located at a definite point on the line. Single stub is adequate for a openwire line. For coaxial lines, placement of stub at exact point is difficult. ii) Two adjustments were required in single stub, these being the location and length of the stub. iii) This technique is suitable for fixed frequency only. If frequency changes, the location of the stub has to be changed. To overcome this difficulty, double stub impedance matching is used. In this system, two different short circuited stubs are used for impedance matching, location of the stubs is arbitrary and the spacing between the two stubs is made $\frac{\lambda}{4}$.	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	The spacing d_2 is frequently made ($\lambda/4$). A spacing of ($3\lambda/8$) is also sometimes used, if the first stub is to be located very near to the load. ($\lambda/2$) spacing is avoided becuase it places the two stubs effectively in parallel at the same point.  Fig. 3.12: Illustrating double stub matching Let the first stub whose length is l_1 be located at the point 1 1' at a distance d_1 from the laod end. Consider a transmission line length l terminated with a load impedance Z_R. Assume that two stubs A and B having a characteristic impedance same as that of the line characteristic impedance are connected in parallel nearer to the main line.	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	Let l_{s_1} and l_{s_2} be the fixed locations of stub A and stub B respectively from the load. Also, let l_{t_1} and l_{t_2} be the lengths of stub A and stub B respectively. Assume that the distance between two stubs is $l_s < \lambda/2$. The design of a double stub matching is different from a single stub matching. Here the locations of the stubs are fixed as shown in Fig. 10.27. Matching is achieved by adjusting the stub lengths such that the admittance at the second stub is equal to the characteristic admittance. To accomplish this, we use the Smith chart as follows. Steps for using Smith chart: 1. Locate the normalised load impedance (z_R) point as P. 2. Locate the normalised load admittance point Q which is opposite P. 3. Draw the S circle with centre O and radius OP. The length OP on the horizontal axis gives the VSWR value before the stub is connected. 4. Move the admittance point Q towards the generator (clockwise direction) to a distance l_{s_1} at point Q_1 on the S circle. This point represents normalised admittance at stub A before stub connection. 5. Draw circle 1 for unity conductance ($y = 1$ or $R = 1$ circle) on the right-hand side of the chart for stub B. 6. Since the distance between the two stubs is l_s, all the points of the unity conductance circle should be shifted by l_s distance towards the load. Draw circle 2 to get the shifted unity conductance circle for stub A. 7. Since stub A adds a susceptance to the load, move point Q_1 along the constant conductance circle to a point C on the circle 2. Point C represents the combined normalised admittance at stub A after stub connection. Change in susceptance = Susceptance at point C − susceptance at point Q_1. 8. The susceptance of the stub A is equal to the change in susceptance, i.e., y_{stubA}. Locate point C′ on the outer circle for y_{stubA}. 9. The length of the short-circuited stub A, l_{t_1}, is calculated from the right side end of chart at point B (i.e., $R = \infty$ on R circle) to the point C′ on the outer circle towards the generator. 10. Draw a circle with centre O and radius OC. It cuts circle 1 at D. Point D gives the admittance on the line at stub B, i.e., y_{s_2} before stub B is connected. 10. To provide proper termination with stub B, the susceptance of the stub B should be equal to the negative value of the susceptance at point D, i.e., y_{stubB}. Locate point D' on the outer circle for y_{stubB}. 12. The length of the short circuited stub B, l_{t_2}, is calculated from the right-side end of the chart at point B (i.e., $R = \infty$ on R circle) to the point D′ on the outer circle towards the generator.	
12	**Explain the principle of impedance matching with quarter wave transformer.**	10

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	A transmission line of length $\frac{\lambda}{4}$ is called the quarter-wave line. The generalised expression for the input impedance of the dissipationless line, $$Z_s = R_0 \left[\frac{Z_R + jR_0 \tan(\beta \cdot s)}{R_0 + jZ_R \tan(\beta \cdot s)}\right]$$ Re-arranging the above equation, $$Z_s = R_0 \left[\frac{\dfrac{Z_R}{\tan\left(\dfrac{2\pi s}{\lambda}\right)} + jR_0}{\dfrac{R_0}{\tan\left(\dfrac{2\pi s}{\lambda}\right)} + jZ_R}\right] \text{ where } \beta = \frac{2\pi}{\lambda} \quad \text{... (3.4)}$$ If the line is made quarter wave long or $s = \frac{\lambda}{4}$ $\frac{2\pi s}{\lambda} = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$, $\tan\left(\frac{\pi}{2}\right) = \infty$ $Z_s = R_0 \left[\frac{jR_0}{jZ_R}\right]$, $\boxed{Z_s = \frac{R_0^2}{Z_R}.}$ ... (3.5)	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	(i.e.,) The input impedance of the line is equal to the square of R_0 of the line divided by the load impedance. A quarter wave section of line may be thought of as a transformer to match a load of Z_R ohms to a source of Z_S ohms. Such a match can be obtained if the characteristic impedance R_0' of the matching quarter wave section of the line is properly chosen according to, $$R_0' = \left\lvert \sqrt{Z_S \cdot Z_R} \right\rvert . \qquad \text{... (3.6)}$$ R_0' of the matching section should be equal to the geometric mean of the source and load impedance.  	

Question No	Topic Learning Outcomes/ Question Bank/ MCQ/ Lab Experiments/ Tutorial Task Labels to the right indicate: CO=Course Outcome, BL=BLOOMS level. On completion of the Lecture/ Tutorial/ Lab on a topic, students must be able to:	Marks per Q
	A quarter wave line may be considered as an impedance inverter in that it can transform a low impedance in to high impedance and vice versa. An important application of the quarter wave matching section is to couple a transmission line to a resistive load such as an antenna. The quarterwave matching section then must be designed to have a characteristic impedance R_0' so chosen that the antenna resistance R_A is transferred to a value equal to the characteristic impedance R_0 of the transmission line. The characteristic impedance of the matching section is then given by, $$R_0' = \sqrt{R_A \cdot R_0} \quad \text{... (3.7)}$$ The range of values of R_A and R_0 that can be matched satisfactorily is limited roughly to about 10 to 1. A quarter wave transformer may also be used if the load is not pure resistance. It should then be connected between points corresponding to I_{max} or E_{min} at which places the transmission line has resistive impedance given by $\frac{R_0}{S}$ or SR_0. For step down in impedance from the line value of R_0 the matching transformer characteristic impedance should be, $$R_0' = \sqrt{R_0 \cdot \frac{R_0}{S}} = R_0\sqrt{\frac{1}{S}} \quad \text{... (3.8)}$$ Another application of the short circuited quarter wave line is an insulator to support an open wire line or the center conductor of a coaxial cable. Such lines are referred to as **copper insulators**.  Fig. 3.2 : Quarterwave line as-insulator	

Faculty Incharge **HOD-ECE**